EDR Introspection

Enabling deep insights into EDRs analysis components

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Introduction

EDR Intro

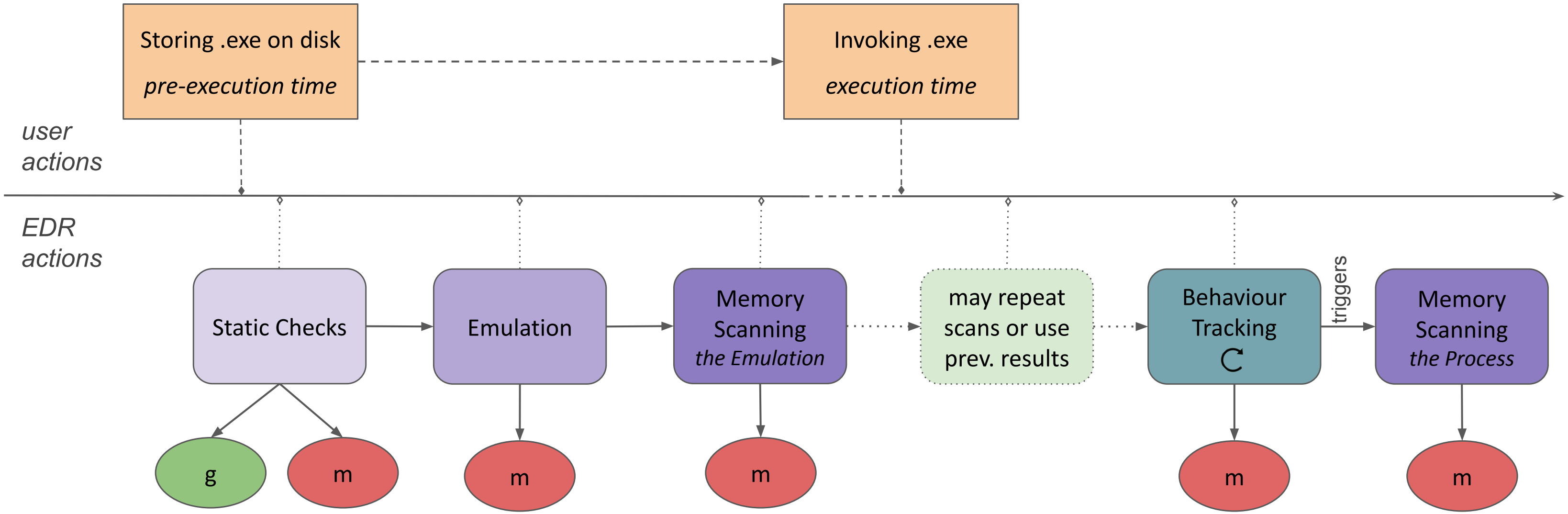
- **E**ndpoint **D**etection and **R**esponse
- EDRs monitor and protect endpoints against attacks
- Static analysis components to detect malware (like AVs) plus
    - Emulation and runtime monitoring of processes
    - Sending telemetry to a centralized server
    - Organisation-wide response possibilities, manual or automatic

EDRs act like black-boxes, vendors do not document the inner workings and shortcomings of their products

Develop a framework to systematically execute attacks and reveal the EDR’s exact actions

Related Work

EDR Overview



EDR’s supervision of processes

- EDRs track process actions via system calls, from:
  - Parsing events from various **ETW traces**
  - **Function hooking** in processes
    - deprecated due to various bypasses

- MDE & Defender
  - write debug info to **Antimalware** and **Threat-Intel** ETW traces for:
    - Actions of other processes
    - **Actions of own process**

Method

Monitoring Sources

How to supervise the EDR itself

- Kernel-mode Monitoring
  - SSDT hooking
  - Hypervisor Hooking

- User-mode Monitoring
  - ETW traces like Kernel and Antimalware
  - ETW-TI trace > *exploits needed*
  - NTDLL hooking > *exploits needed*

User-mode sources

- Kernel ETW traces
  - Process start stop, open process requests, creating and opening files, new network connections
- Antimalware-Engine ETW trace
  - Debug info about the EDRs actions

- ETW-TI trace
  - Summary of relevant actions of attacks actions
  - *Normally no access*
- Low-Level EDR Telemetry
  - Gain visibility over each OpenProcess, ReadMemory, ReadFile, etc. by the EDR
  - *Normally no access*

EDR Protections

MDE as an Example

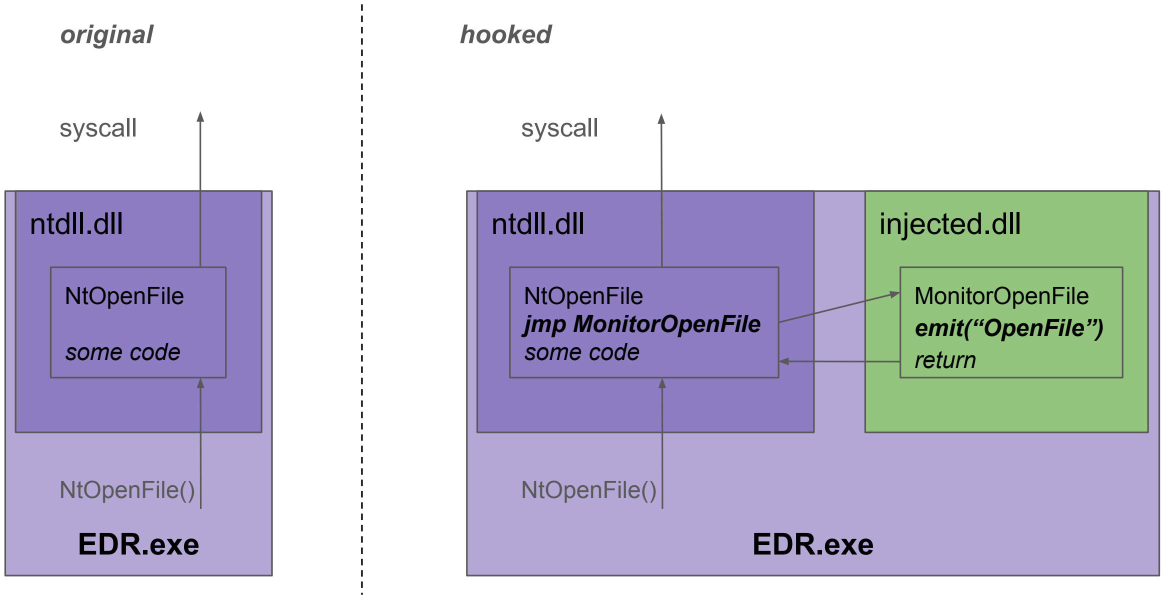
- EDRs run as **PPL processes**
  - Only other PPL processes have access to MDE
- **Restricted Process Handles**
  - Kernel callbacks restrict full access to MDE, even with PPL
- **Restricted DLL loading**
  - Only MS-signed DLLs can be loaded into MDE, even with full access

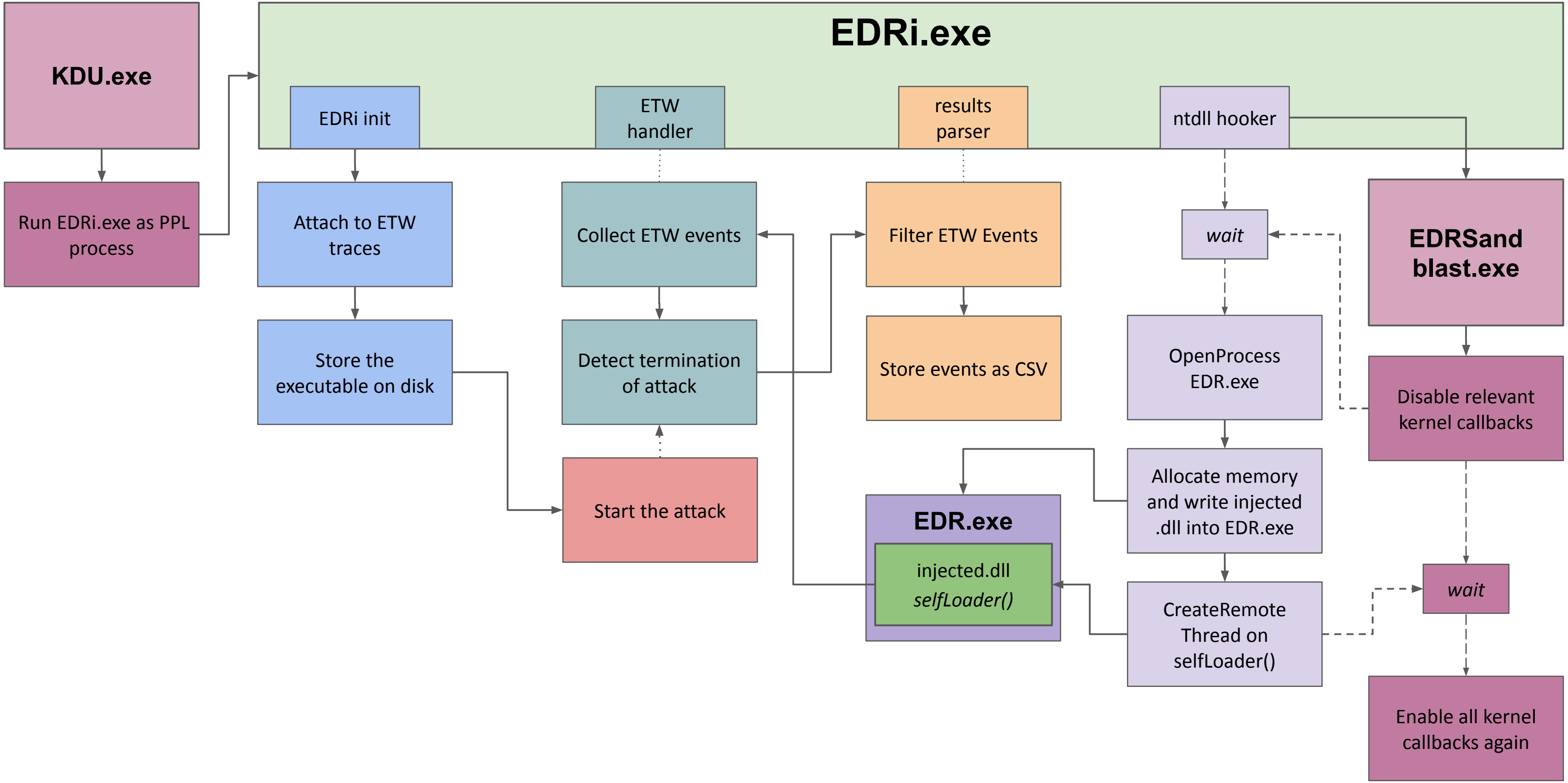
Bypassing EDR Protections

MDE as an Example

- **KDU** to run monitoring process as PPL
  - Monitoring process can now call **OpenProcess(EDR.exe)**
- **EDRSandblast** to temporarily remove kernel callbacks
  - Monitoring process can now call OpenProcess(EDR.exe) with **full access rights**
- **Reflective DLL injection** to insert code into ntdll.dll
  - Monitoring process can now **hook functions** and monitor the EDR.exe

Hooking the EDR

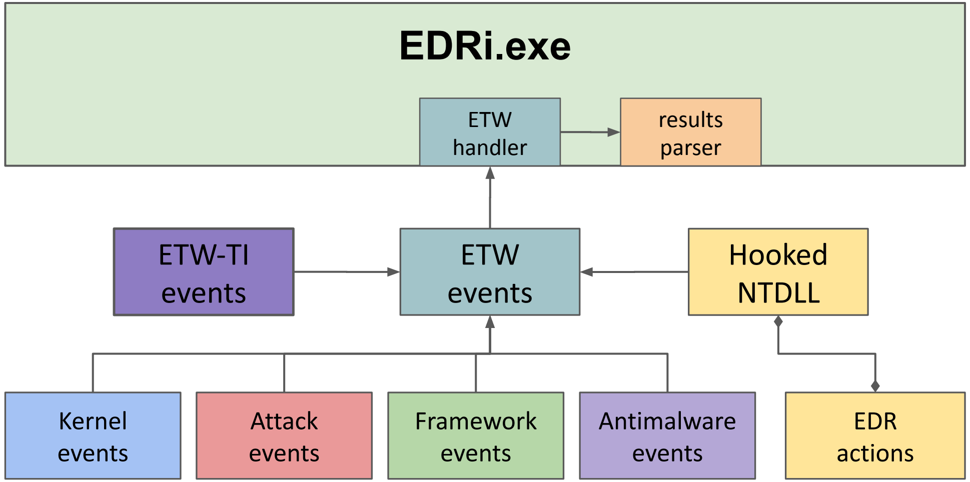




KDU.exe
Run EDRi.exe as PPL process
EDRi.exe
EDRi init
ETW handler
results parser
ntdll hooker
Attach to ETW traces
Store the executable on disk
Collect ETW events
Detect termination of attack
Start the attack
Filter ETW Events
Store events as CSV
wait
OpenProcess EDR.exe
Allocate memory and write injected .dll into EDR.exe
CreateRemote Thread on selfLoader()
EDR.exe
injected.dll
selfLoader()
EDRSandblast.exe
Disable relevant kernel callbacks
wait
Enable all kernel callbacks again

Summary

- automatic start/stop
- 6 event sources
- 3 attacks * 4 variants



Results

Attacks & Events

Event Statistics

MDE vs ...

Attack	Data Source				Total
	Antimalware	Threat Intel	Kernel	EDR Monitor	
Proc Inject	12973	183	4942	5712	23834
C2 Loader	4695	728	3375	10569	19386
Lsass Read	3287	2598	2204	7983	16096

Event Statistics - Filtered

MDE vs ...

Attack	Data Source				Total
	Antimalware	Threat Intel	Kernel	EDR Monitor	
Proc Inject	855	21	132	165	1197
C2 Loader	153	24	288	1283	1767
Lsass Read	1526	8	53	169	1780

Results

[MDE vs Lsass Reader]

Storing the Attack

task_info	process_id	pic	tar	message	filepath
EDRiTask	11224 EDRi.exe			++ EDRi START MARKER ++	
CreateNewFile Info	4 System				C:\Windows\System32\LogFiles\WMI\RtBackup\EtwRTEDRi-Misc.etl
NameCreate Info	4 System				C:\Windows\System32\LogFiles\WMI\RtBackup\EtwRTEDRi-Misc.etl
EDRiTask	11224 EDRi.exe			Before decrypting the attack exe from C:\Users\admin3\Desktop\EDR-Int	
CreateNewFile Info	11224 EDRi.exe				C:\Users\Public\Downloads\attack-319.exe
NameCreate Info	11224 EDRi.exe				C:\Users\Public\Downloads\attack-319.exe
Stream scan request Start	5256 MsMpEng.exe	11224			C:\Users\Public\Downloads\attack-319.exe
EDRiTask	11224 EDRi.exe			After decrypting the attack exe	
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
Message Info	5256 MsMpEng.exe			subtype=Lowfi, sigseq=0x00000555D4FB9C24, sigsha=68b7f163c37677d744a8	
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe				
Message Info	5256 MsMpEng.exe			subtype=Lowfi, sigseq=0x000060E78EC7B1CB, sigsha=88d8d1535ab7761f8775	
Stream scan request Stop	5256 MsMpEng.exe	11224			C:\Users\Public\Downloads\attack-319.exe

Store attack

Static checks with MetaStore

Attack Startup

task_info	process_id	pic	tar	message	filepath	cachename	result
ProcessStart Start	11064 powershell.e		1132		C:\Users\Public\Downloads\attack-319.exe		
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire			
Cache MOACLookup	5256 MsMpEng.exe				C:\Users\Public\Downloads\attack-319.exe		0x8002
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire			
Cache MOACLookup	5256 MsMpEng.exe				C:\Users\Public\Downloads\attack-319.exe		0x8002
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire			
Cache MOACLookup	5256 MsMpEng.exe				C:\Users\Public\Downloads\attack-319.exe		0x8002
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire			
Cache CacheLookup	5256 MsMpEng.exe				C:\Users\Public\Downloads\attack-319.exe	USN Cache	MISS
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire			
HookTask	5256 MsMpEng.exe		1132	NtOpenProcess with 0x600:PROCESS_SET_INFORMATION|PROCESS_QUERY_INFORM			
PsOpenProcess	5256 MsMpEng.exe		1132				
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessLoggingInformation			
HookTask	5256 MsMpEng.exe		1132	NtClose process			
Behavior Monitoring ProcessMonito	5256 MsMpEng.exe	1132			C:\Users\Public\Downloads\attack-319.exe		
Behavior Monitoring BmProcessCrea	5256 MsMpEng.exe	1132			C:\Users\Public\Downloads\attack-319.exe		
HookTask	5256 MsMpEng.exe		1132	NtOpenProcess with 0x1000:PROCESS_QUERY_LIMITED_INFORMATION			
PsOpenProcess	5256 MsMpEng.exe		1132				
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessSessionInformation			
HookTask	5256 MsMpEng.exe		1132	NtClose process			
Behavior Monitoring BmProcessConte	5256 MsMpEng.exe	1132					
Behavior Monitoring BmInternal	5256 MsMpEng.exe	1132					
ThreadStart Start	11064 powershell.e		1132				

Attack start

Static checks with MOAC

Get logging & session information

Attack Startup

task_info	process_id	pic	tar	message	filepath	cachename	result	action
PsOpenThread	5116 csrss.exe		1132					
HookTask	5256 MsMpEng.exe		5256	NtOpenFile C:\Users\Public\Downloads\attack-319.exe with DesiredAcces				
ImageLoad Info	1132 attack-319.ex		1132	(first load)	C:\Windows\System32\ntdll.dll			
ExpensiveOperationTask ExpensiveOpe	5256 MsMpEng.exe			BM InitializeFriendlyStatus::IsKnownFriendly()	C:\Users\Public\Downloads\attack-319.exe			
ExpensiveOperationTask ExpensiveOpe	5256 MsMpEng.exe			BM InitializeFriendlyStatus::IsKnownFriendly()	C:\Users\Public\Downloads\attack-319.exe			
HookTask	5256 MsMpEng.exe		1132	NtOpenProcess with 0x1000:PROCESS_QUERY_LIMITED_INFORMATION				
PsOpenProcess	5256 MsMpEng.exe		1132					
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessSessionInformation				
HookTask	5256 MsMpEng.exe		1132	NtClose process				
HookTask	5256 MsMpEng.exe		1132	NtOpenProcess with 0x1000:PROCESS_QUERY_LIMITED_INFORMATION				
PsOpenProcess	5256 MsMpEng.exe		1132					
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessSessionInformation				
HookTask	5256 MsMpEng.exe		1132	NtClose process				
Behavior Monitoring BmOpenProcess	5256 MsMpEng.exe	11064	1132					
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe						0	query
Behavior Monitoring BmInternal	5256 MsMpEng.exe	1132						
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe						0	query
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe						0	query
PsOpenProcess	9584 conhost.exe		1132					
Behavior Monitoring BmOpenProcess	5256 MsMpEng.exe	9584	1132					
MetaStoreTask MetaStoreAction	5256 MsMpEng.exe						0	query

Attack loads NTDLL

More static checks with MetaStore

Attack Startup

task_info	process_id	pic	tar	message	filepath	cachename	result	action
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire				
HookTask	5256 MsMpEng.exe		1132	NtOpenProcess with 0x1410:PROCESS_VM_READ|PROCESS_QUERY_INFORMATION|P				
PsOpenProcess	5256 MsMpEng.exe		1132					
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessSessionInformation				
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessBasicInformation				
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x8 bytes from (NoImage) Private!0x0000001bfd58e0				
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x8 bytes from (NoImage) Private!0x0000001bfd58e0				
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x8 bytes from (PEB) C:\WINDOWS\SYSTEM32\ntdll.dl				
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x88 bytes from (NoImage) Private!0x00000139fc433				
HookTask	5256 MsMpEng.exe		1132	NtClose process				
Cache MOACLookup	5256 MsMpEng.exe				C:\Windows\System32\ntdll.dll		0x8001	
PsOpenProcess	1132 attack-319.e		1132					
ThreadStart Start	1132 attack-319.e		1132					
ThreadStart Start	1132 attack-319.e		1132					
ThreadStart Start	1132 attack-319.e		1132					
ImageLoad Info	1132 attack-319.e		1132	(last load)	C:\Windows\System32\msvcp140.dll			
Cache MOACLookup	5256 MsMpEng.exe				C:\Windows\System32\msvcp140.dll		0x8001	
AttackTask	1132 attack-319.e			Reader started with PID 1132				
PsOpenProcess	3592 MsSense.exe		1132					
PsOpenProcess	3592 MsSense.exe		1132					
Scan request Start	5256 MsMpEng.exe		1132		pid:1132			
HookTask	5256 MsMpEng.exe		1132	NtOpenProcess with 0x101000:PROCESS_QUERY_LIMITED_INFORMATION|SYNCHRO				
PsOpenProcess	5256 MsMpEng.exe		1132					
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessSessionInformation				
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessImageFileName				
HookTask	5256 MsMpEng.exe		1132	NtClose process				
UfsScanFileTask Start	5256 MsMpEng.exe				C:\Users\Public\Downloads\attack-319.exe			
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire				
HookTask	5256 MsMpEng.exe		5256	NtCreateFile \??\C:\Users\Public\Downloads\attack-319.exe with Desire				
HookTask	5256 MsMpEng.exe		5256	NtReadFile C:\Users\Public\Downloads\attack-319.exe				
KERNEL_THREATINT_TASK_ALLOCVM	5256 MsMpEng.exe		5256					
KERNEL_THREATINT_TASK_PROTECTVM	5256 MsMpEng.exe		5256					
KERNEL_THREATINT_TASK_PROTECTVM	5256 MsMpEng.exe		5256					
HookTask	5256 MsMpEng.exe		5256	NtReadFile C:\Users\Public\Downloads\attack-319.exe				

Read attack's PEB

First output of the attack

Emulation

Attack Runtime

task_info	process_id	pic	tar	message	filepath
UfsScanProcTask Start	5256 MsMpEng.exe	1132			\.\proc\Process:1132,134102806531128903
HookTask	5256 MsMpEng.exe		1132	NtOpenProcess with 0x418:PROCESS_VM_OPERATION|PROCESS_VM_READ|PROCESS_	
PsOpenProcess	5256 MsMpEng.exe		1132		
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessSessionInformation	
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessBasicInformation	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x8 bytes from (NoImage) Private!0x0000001bfd58e0	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x8 bytes from (PEB) C:\WINDOWS\SYSTEM32\ntdll.dl	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc43	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc43	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc43	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc43	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc43	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc44	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc44	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc43	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc44	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc44	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc44	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc44	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x00000139fc44	
HookTask	5256 MsMpEng.exe		1132	NtQueryInformationProcess with InfoClass=ProcessBasicInformation	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x8 bytes from (NoImage) Private!0x0000001bfd58e0	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x8 bytes from (PEB) C:\WINDOWS\SYSTEM32\ntdll.dl	
HookTask	5256 MsMpEng.exe		1132	NtReadVirtualMemory 0x88 bytes from (NoImage) Private!0x00000139fc433	

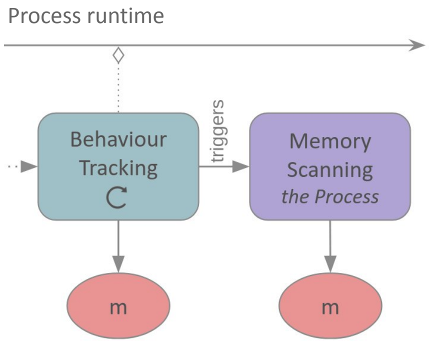
Read attack's loaded modules + PEB

Conclusion

New Deconditioning Bypass

Lsass Process Dumping

- Exhaust behavioural triggers for **process dumping**
  - Dump 20 non-important processes, then lsass
- Existing deconditioning approaches
  - **[SirAllocALot]**: allocate and execute benign code, 10k times (2024)
  - **[BamboozlEDR]**: inject benign events into ETW to fill traces (2025)
- Generalised deconditioning theory (new)
  - Find the trigger & exhaust it with benign events ⇒ runtime bypass
  - Time and space constraints also apply to EDRs



EDR Introspection

- Chain 3 exploits to gain full control ⟶ visibility
  - PPL, kernel callback disabling, reflective DLL injection
- Extensive visibility over all EDR actions
  - Supported by various existing ETW traces
- Confirms suspected EDR mechanisms
  - **Static Identifiers**
  - **Emulation Engine**
  - **Runtime Triggers**

- No visibility over static identifiers and inside emulation engine
  - Exists in isolation in [t-tani/defender2yara] and [0xAlexei/mpclient]

Discussion

- Future role of EDRs
  - Various bypasses found, even when not the main focus
  - Shift to behaviour and baseline-driven detections [MITRE]
- Future research
  - More data sources → deeper understanding
  - EDRi Core, optimized for malware development
  - Introspect more EDRs with this approach

Questions ?

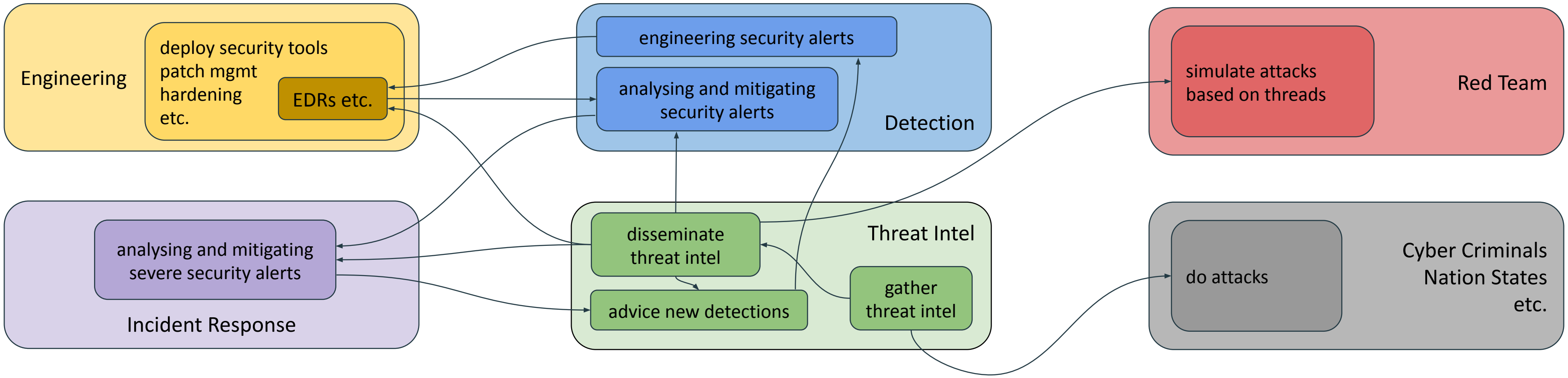
Backup

Objectives

EDRs act like black-boxes, vendors do not document the inner workings and shortcomings of their products.

- Explain techniques EDRs how analyse executables, and investigate common bypasses.
- Enumerate existing techniques to enable supervision of the EDRs actions.
- Create a framework to systematically execute attacks and monitor the EDRs actions.
- Evaluate the results and reveal the EDRs actions based on the attacks.

Modern Security Environment



EDR Detection Mechanisms

“the only general way to obtain information on the behavior of a program is to run the program” [1]
- Rice-Shapiro

- Static Detection Identifiers
  - Hashes, Signatures, YARA rules
  - Extracted from disk, emulation or memory
  - Some public databases of known bad
- Behaviour-based Triggers
  - SIGMA & KQL
  - Runtime tracking via **ntdll hooks** (obsolete) or **Event Tracing for Windows (ETW)** events
  - Some public databases of known bad

EDR Protections

- EDRs run as PPL processes, only other PPL processes have access
  - **Kernel-mode exploits** (KDU) to run any process as PPL
- Kernel callback handle downgrading
  - myProcess.exe OpenProcess(explorer.exe) → all rights
  - myProcess.exe OpenProcess(EDR.exe) → restricted rights
  - **Kernel-mode exploits** (EDRSandblast) to disable this restrictions
- Only MS-signed DLLs can be loaded into MsMpEng.exe
  - **Reflective DLL injection** bypasses these checks

Kernel Callback Disabling

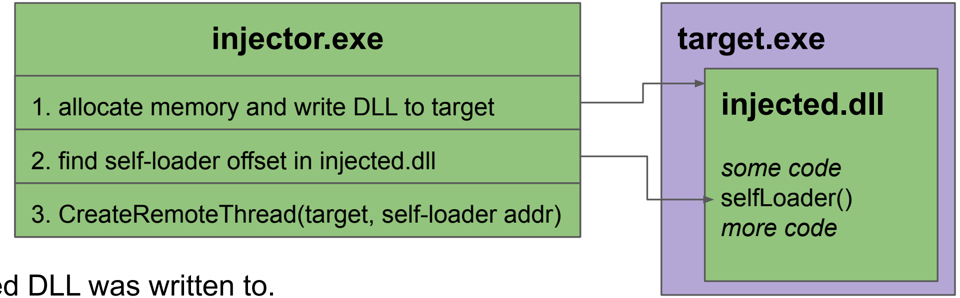
```
PS C:\Users\hacker\source\repos\myEDRSandblast\x64\Release> .\EDRSandblast.exe toggle_callbacks 01 --kernelmode -i

 D3FC0N 30 Edition | Maxime MEIGNAN (@th3m4ks) & Thomas DIOT (@_Qazeer)

[!] If kernel mode bypass is enabled, it is recommended to enable usermode bypass as well (e.g. to unhook the NtLoadDriver API call)

[===== KERNEL MODE =====]

[+] Loading required offsets for ntoskrnl.exe...
[+] Downloading kernel related offsets from the MS Symbol Server (will drop a .pdb file in current directory)
[+] Downloading offsets succeeded !
[+] Downloading fltmgr.sys related offsets from the MS Symbol Server (will drop a .pdb file in current directory)
[+] Downloading offsets succeeded !
[+] Installing vulnerable driver...
[*] 'flL5Kz4n' service was not present
[+] 'flL5Kz4n' service is successfully registered
[+] 'flL5Kz4n' service ACL configured to for Everyone
[+] 'flL5Kz4n' service started


[+] Checking if any EDR kernel notify rountines are set for image loading, process and thread creations...
[+] Checking if EDR callbacks are registered on processes and threads handle creation/duplication...
[+] [ObjectCallblacks]  Enumerating Process object callbacks :
[+] [ObjectCallblacks]          Callback at FFFFD20B274DD830 for handle creations:
[+] [ObjectCallblacks]                  Status: Enabled
[+] [ObjectCallblacks]                  Preoperation at 0xfffff80135ba4830 [UCPD.sys + 0x4830]
[+] [ObjectCallblacks]          Callback at FFFFD20B2746C440 for handle creations & duplications:
[+] [ObjectCallblacks]                  Status: Enabled
[+] [ObjectCallblacks]                  Preoperation at 0xfffff801333ce540 [WdFilter.sys + 0x2e540]
[+] [ObjectCallblacks]                  Callback belongs to an EDR and is enabled!
[+] [ObjectCallblacks]  Enumerating Thread object callbacks :
[+] [ObjectCallblacks]          Callback at FFFFD20B274DD870 for handle creations:
[+] [ObjectCallblacks]                  Status: Enabled
[+] [ObjectCallblacks]                  Preoperation at 0xfffff80135ba4830 [UCPD.sys + 0x4830]
[+] [ObjectCallblacks]  Object callbacks are present !
[+] Disabling all EDR callbacks...
[+] [ObjectCallblacks]  Disabling WdFilter.sys callback...

[+] Press ENTER to enable callbacks again:

[*] Re-enabling all EDR callbacks...
[+] [ObjectCallblacks]  Enabling WdFilter.sys callback...
[*] Uninstalling vulnerable driver...
[+] 'flL5Kz4n' service stopped
[+] The vulnerable driver was successfully uninstalled!
PS C:\Users\hacker\source\repos\myEDRSandblast\x64\Release>
```

```
PS C:\Users\hacker\source\repos\EDR-Introspection> C:\Users\hacker\source\repos\EDR-Introspection\h
elpers\KDU\kdu.exe -pse "C:\Users\hacker\source\repos\EDR-Introspection\x64\Release\HookTester.exe
3232 C:\Users\hacker\source\repos\EDR-Introspection\x64\Release\Hooks.dll" -prv 54
```

```
[+] Kernel memory read at FFFFC005085F767A succeeded
        PsProtection->Type: 0 (PsProtectedTypeNone)
        PsProtection->Signer: 0 (PsProtectedSignerNone)
        PsProtection->Audit: 0
[+] Process object modified
[+] Kernel memory read at FFFFC005085F767A succeeded
        New PsProtection: 0x31
        PsProtection->Type: 1 (PsProtectedTypeProtectedLight)
        PsProtection->Signer: 3 (PsProtectedSignerAntimalware)
        PsProtection->Audit: 0
[+] Hooker: GrantedAccess: 0x1fffff -> full access
[*] HookTester: DLL injection succeeded.
```

```
[+] Kernel memory read at FFFFC004FD73C67A succeeded
        PsProtection->Type: 0 (PsProtectedTypeNone)
        PsProtection->Signer: 0 (PsProtectedSignerNone)
        PsProtection->Audit: 0
[+] Process object modified
[+] Kernel memory read at FFFFC004FD73C67A succeeded
        New PsProtection: 0x31
        PsProtection->Type: 1 (PsProtectedTypeProtectedLight)
        PsProtection->Signer: 3 (PsProtectedSignerAntimalware)
        PsProtection->Audit: 0
[+] Hooker: GrantedAccess: 0x1ff7d4 -> PROCESS_SET_SESSIONID | PROC
PROCESS_SET_LIMITED_INFORMATION, not including: PROCESS_TERMINATE |
```

Reflective DLL Injection



1. Get the base address where the injected DLL was written to.
2. Find where kernel32.dll is loaded in the target process via traversing the PEB.
3. Get important function addresses from kernel32.dll via traversing its export table.
4. Allocate memory for all imported DLLs used by injected.dll.
5. For each imported DLL, copy their sections and resolve their import address tables.
6. Process all relocations, i.e. correct the dummy addresses with real addresses.
7. Call the actual injected.dll's entrypoint.

Attacks & Detection Rates

Strategy > v Attack	standard	obfuscation	obfuscation + anti-emulation	obfuscation + anti-emulation + deconditioning
Process Injection	*not detected on the first run*	*not detected*	*alerted when executed*	*not detected*
Lsass Reader	*freezes when dumping lsass*	*freezes when dumping lsass*	*freezes when dumping lsass*	*alerted when dumping lsass*
CobaltStrike Loader	*not detected*	*blocked when stored*	*not detected*	*not detected*

task_info	process_id	tar	message	filepath
EDRiTask	3536 EDRi.exe		++ EDRi START MARKER ++	
CreateNewFile Info	4 System			C:\Windows\System32\LogFiles\WMI\RtBackup\EtwRTEDRi-Misc.etl
NameCreate Info	4 System			C:\Windows\System32\LogFiles\WMI\RtBackup\EtwRTEDRi-Misc.etl
EDRiTask	3536 EDRi.exe		Before decrypting the attack exe from C:\Users\admin\Desktop\EDR-Introspection\	
CreateNewFile Info	3536 EDRi.exe			C:\Users\Public\Downloads\attack-369.exe
NameCreate Info	3536 EDRi.exe			C:\Users\Public\Downloads\attack-369.exe
Stream scan request Start	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
EDRiTask	3536 EDRi.exe		After decrypting the attack exe	
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
Message Info	8892 MsMpEng.exe		subtype=Lowfi, sigseq=0x00000555D4FB9C24, sigsha=68b7f163c37677d744a8a0d9772ac4	
ExpensiveOperationTask Exp	8892 MsMpEng.exe		LowFi	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		LowFi	C:\Users\Public\Downloads\attack-369.exe
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
MetaStoreTask MetaStoreAct	8892 MsMpEng.exe			
Message Info	8892 MsMpEng.exe		subtype=Lowfi, sigseq=0x000060E78EC7B1CB, sigsha=88d8d1535ab7761f877525fe19896	
ExpensiveOperationTask Exp	8892 MsMpEng.exe		SyncLowFi	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		SyncLowFi	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		GetHashes(GetCurrentFileHash)	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		GetHashes(GetCurrentFileHash)	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		IsSignedFile	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		GetHashes	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		GetHashes	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		GetHashes(ValidateTrust)	C:\Users\Public\Downloads\attack-369.exe
ExpensiveOperationTask Exp	8892 MsMpEng.exe		GetHashes(ValidateTrust)	C:\Users\Public\Downloads\attack-369.exe
Message Info	8892 MsMpEng.exe		C:\Users\Public\Downloads\attack-369.exe is NOT trusted. Checking Trust? 1	
Message Info	8892 MsMpEng.exe		IsSignedFile failed on C:\Users\Public\Downloads\attack-369.exe (CheckTrust=1)	
ExpensiveOperationTask Exp	8892 MsMpEng.exe		IsSignedFile	C:\Users\Public\Downloads\attack-369.exe
CreateNewFile Info	8892 MsMpEng.exe			C:\ProgramData\Microsoft\Windows Defender\Scans\History\Store\2
NameCreate Info	8892 MsMpEng.exe			C:\ProgramData\Microsoft\Windows Defender\Scans\History\Store\2
Stream scan request Stop	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe

Start EDRi

Store Attack

Static Checks

task_info	process_id	tar	message	filepath
AttackTask	14268 attack-369.exe		Doing anti emulation calc operations for about 5 sec	
EDRiTask	3536 EDRi.exe		After starting the attack exe	
HookTask	8892 MsMpEng.exe	14268	NtOpenProcess with 0x101000:PROCESS_QUERY_LIMITED_INFORMATION|SYNCHRONIZE	
PsOpenProcess	8892 MsMpEng.exe	14268		
HookTask	8892 MsMpEng.exe	14268	NtQueryInformationProcess with InfoClass=ProcessSessionInfor	
HookTask	8892 MsMpEng.exe	14268	NtQueryInformationProcess with InfoClass=ProcessImageFileName	
HookTask	8892 MsMpEng.exe	14268	NtClose process	
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache MOACLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache MOACLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache MOACLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache CacheLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache MOACLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache MOACLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache MOACLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
HookTask	8892 MsMpEng.exe	8892	NtCreateFile \??\C:\Users\Public\Downloads\attack-369.exe with DesiredAccess=0x100080:F	
Cache CacheLookup	8892 MsMpEng.exe			C:\Users\Public\Downloads\attack-369.exe
UfsScanProcTask Start	8892 MsMpEng.exe			\.\proc\Process:14268,134108824977290439
HookTask	8892 MsMpEng.exe	14268	NtOpenProcess with 0x418:PROCESS_VM_OPERATION|PROCESS_VM_READ|PROCESS_QUERY_INFORMATION	
PsOpenProcess	8892 MsMpEng.exe	14268		
HookTask	8892 MsMpEng.exe	14268	NtQueryInformationProcess with InfoClass=ProcessSessionInformation	
HookTask	8892 MsMpEng.exe	14268	NtQueryInformationProcess with InfoClass=ProcessBasicInformation	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x8 bytes from (NoImage) Private!0x000000bd24922018	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x8 bytes from (PEB) C:\WINDOWS\SYSTEM32\ntdll.dll!0x00007ffeadff2	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372936a0	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f237293470	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f237298410	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f237298ac0	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f237298760	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372a0660	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372a0ac0	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f23729de30	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372a1220	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372a1de0	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372a1880	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372a15e0	
HookTask	8892 MsMpEng.exe	14268	NtReadVirtualMemory 0x138 bytes from (NoImage) Private!0x000001f2372a1b20	

Start Attack

Check Start

Emulation

Memory Scans

Event Interpretation

Event	Assumed Objective
stream scan	Start and stop the static analysis phase, gets static identifiers
scan	Start and stop the emulation (engine), gets dynamic identifiers
UfsScanFile/Proc	The actual emulation, based on a file on disk / process memory
SpyNet report	Bundles identifiers, submits them to Microsoft’s cloud
“Defender folder” access	Past detections, logging scan results, compiling detection data or quarantining malware

Events Mapping (extensive)

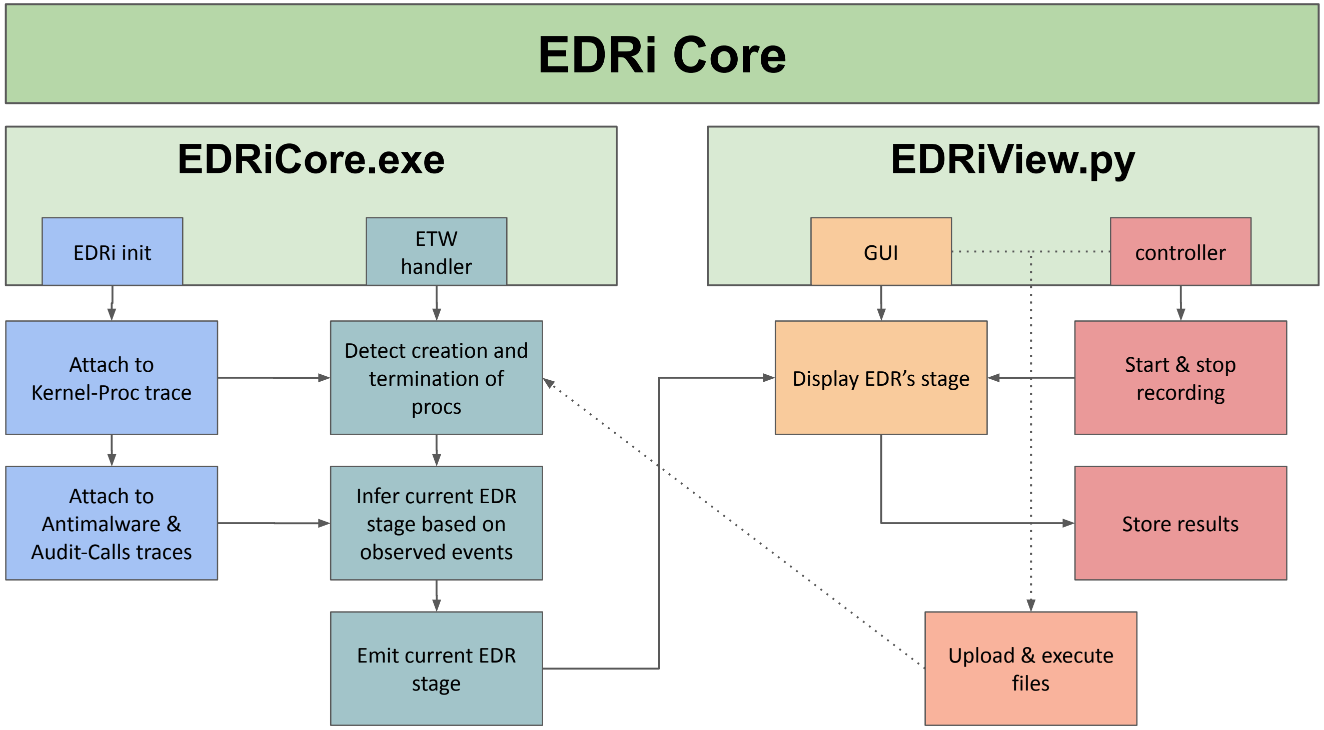
Event	**Assumed Objective(s)**
stream scan	Start and stop the static analysis phase, gets static identifiers
MetaStore insert	Stores extracted identifiers from static scans and emulations
MetaStore query	Gets identifiers, used to decide what further analysis must be done
scan	Start and stop the emulation (engine), gets dynamic identifiers
MOAC lookup	Used to check if a file should be emulated or not
UfsScanFile	The actual emulation of the file
GetHashes	Extracts identifiers (hashes) for a static scan and after an emulation
UfsScanProc	Scans the loaded modules in a process
SpyNet report	Bundles identifiers, submits them to Microsoft’s cloud to request a verdict
“Defender folder” access	Read past detections, logging scan results or moving malware to quarantine

Process Open Interpretation

Process Open Rights *	Observed Time	Assumed Objective
0x410	After behaviour trigger	Extensive memory scan
0x418	Process start	Get loaded modules via LDR
0x600	Process start	Get process' logging information
0x1410	Inconsistent	Get first loaded module (own module)
0x101000	Around emulation	Get the executables file path and name

EDR Detection Stages

1. The identifiers extracted via **static analysis** are known to be malicious.
2. The identifiers found in an **emulation** run are known to be malicious.
3. The attack acts suspiciously at **runtime**, triggering further (memory) scans.


EDRi Core
EDRiCore.exe
EDRi init
ETW handler
Attach to Kernel-Proc trace
Detect creation and termination of procs
Attach to Antimalware & Audit-Calls traces
Infer current EDR stage based on observed events
Emit current EDR stage
EDRiView.py
GUI
controller
Display EDR’s stage
Start & stop recording
Store results
Upload & execute files